COURSE NAME & CODE: ECA 0303 – Signals and systems

NAME: D S Lakshmi Narayana

REG NO: 192512371

Experiment 2.1

Periodic and Aperiodic Signals

Aim

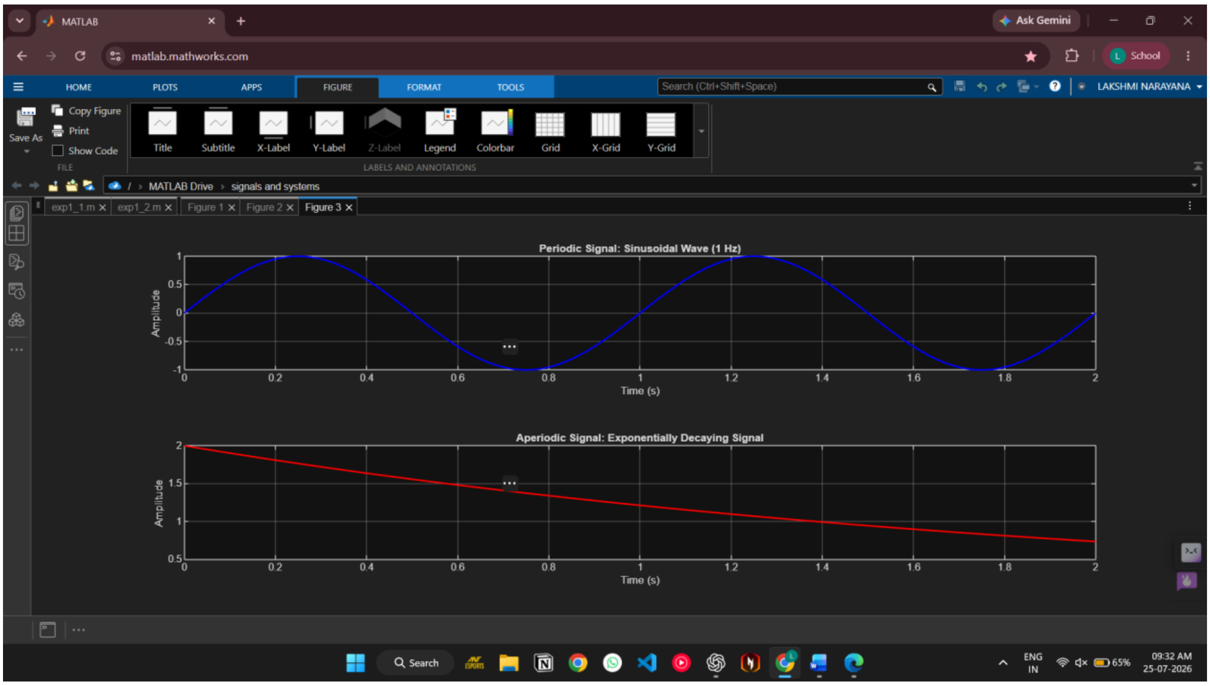
To generate and distinguish between periodic and aperiodic signals using MATLAB. The Aim is to demonstrate the fundamental difference between periodic and aperiodic signals by creating examples of each and analyzing their properties.

Program:

```
% Periodic and Aperiodic Signals
clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Periodic Signal (Sine Wave)
x1 = sin(2*pi*t);
% Aperiodic Signal (Exponential Decay)
x2 = exp(-abs(t));
% Plotting
figure;
subplot(2,1,1);
plot(t,x1,'LineWidth',2);
grid on;
```

```
title('Periodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,x2,'LineWidth',2);
grid on;
title('Aperiodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

Output Waveform:



Result:

- **Periodic Signal:** The sinusoidal signal repeats itself at regular intervals, making it periodic.

- **Aperiodic Signal:** The exponentially decaying signal does not repeat, showcasing that it is aperiodic.

This test case illustrates the key differences between periodic and aperiodic signals by generating and visually analysing both types using MATLAB.